

Score: / 10

Calculus I (MATH 2200)

Spring 2020

Wednesday, 03/11/2020

Quiz 7: Section 3.8, 3.9, 3.10

Name (Print): _____

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Section 3.8

3 pts

1. Find y' given that y satisfies the equation $\sin(xy) = x^2 + y$.

$$\frac{d}{dx}(\sin(xy)) = \frac{d}{dx}(x^2 + y)$$

$$\cos(xy)\left(y + x\frac{dy}{dx}\right) = 2x + \frac{dy}{dx}$$

$$y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 2x + \frac{dy}{dx}$$

$$x\cos(xy)\frac{dy}{dx} - \frac{dy}{dx} = 2x - y\cos(xy)$$

$$\frac{dy}{dx}(x\cos(xy) - 1) = 2x - y\cos(xy)$$

$$\frac{dy}{dx} = \frac{2x - y\cos(xy)}{x\cos(xy) - 1}$$

$$\text{So, } y' = \frac{2x - y\cos(xy)}{x\cos(xy) - 1}.$$

Section 3.9

3 pts

1. Find the derivative of
- $g(t) = 108(2^{t/12})$
- .

By the chain rule and the formula for the derivative of b^x , we find:

$$\begin{aligned} g'(t) &= 108 \frac{d}{dt} (2^{t/12}) \\ &= 108 \ln(2) (2^{t/12}) \left(\frac{1}{12} \right) \\ &= 9 \ln(2) (2^{t/12}). \end{aligned}$$

1 (bonus)

2. Find the derivative of
- $f(x) = x^{\sin(x)}$
- .

First, let us note that $b^x = e^{x \ln(b)}$. So, using this, we may rewrite f as

$$f(x) = x^{\sin(x)} = e^{\sin(x) \ln(x)}.$$

So, by the chain rule,

$$\begin{aligned} f'(x) &= \frac{d}{dx} [\sin(x) \ln(x)] \\ &= e^{\sin(x) \ln(x)} \frac{d}{dx} [\sin(x) \ln(x)] \\ &= e^{\sin(x) \ln(x)} \left(\frac{\sin(x)}{x} + \cos(x) \ln(x) \right) \\ &= x^{\sin(x)} \left(\frac{\sin(x)}{x} + \cos(x) \ln(x) \right). \end{aligned}$$

Section 3.10

2 pts

1. The function $f(x) = \sqrt{x} + x^2 + 1$ is one-to-one for $x \geq 0$ and has an inverse on that interval. Find the slope of the curve $y = f^{-1}(x)$ at the point $(3, 1)$.

The point $(1, 3)$ is on the graph of f ; therefore, $(3, 1)$ is on the graph of f^{-1} . In this case, the slope of the curve $y = f^{-1}(x)$ at the point $(3, 1)$ is the reciprocal of the slope of the curve $y = f(x)$ at $(1, 3)$. Note that

$$f'(x) = \frac{1}{2\sqrt{x}} + 2x$$

which means that $f'(1) = \frac{1}{2} + 2 = \frac{5}{2}$. Therefore,

$$(f^{-1})'(3) = \frac{1}{f'(1)} = \frac{1}{5/2} = \frac{2}{5}.$$

Observe that it is not necessary to find a formula for f^{-1} to evaluate its derivative at a point.

2 pts

2. Find the equation of the line tangent to the graph of $g(x) = \sec^{-1}(2x)$ at the point $(1, \pi/3)$.

The slope of the tangent line at $(1, \pi/3)$ is $g'(1)$. Using the Chain Rule, we have

$$\begin{aligned} g'(x) &= \frac{d}{dx}(\sec^{-1}(2x)) \\ &= \frac{2}{|2x|\sqrt{4x^2 - 1}} \\ &= \frac{1}{|x|\sqrt{4x^2 - 1}}. \end{aligned}$$

So, $g'(1) = 1/\sqrt{3}$, and the equation of the tangent line is

$$y - \frac{\pi}{3} = \frac{1}{\sqrt{3}}(x - 1).$$